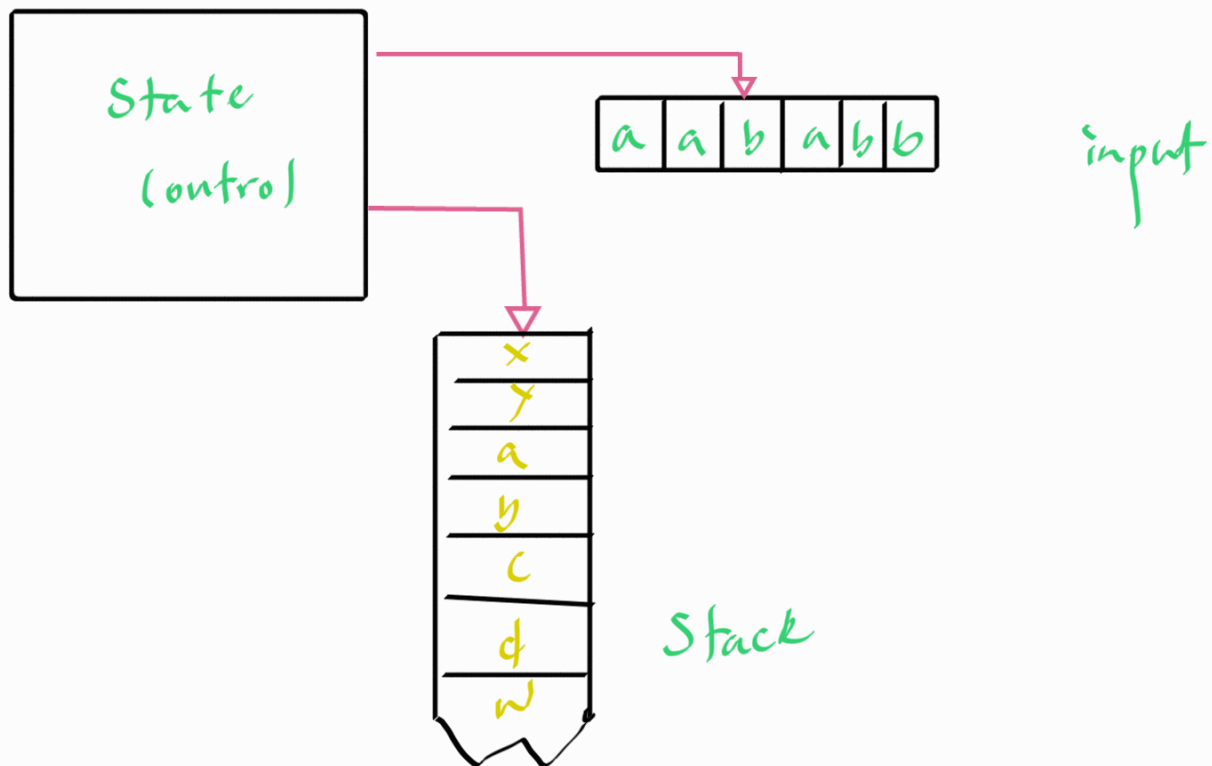


Pushdown Automaton (PDA)

- PDA is basically an NFA + stack.
- Stack provides an additional memory beyond the # of states (memory available)
- PDA is equivalent in power to context free grammar.



$$A = \{0^n 1^n \mid n \geq 0\}$$

Rough idea

~~0~~ ~~0~~ ~~0~~ ~~0~~ ~~0~~ ... ~~0~~ 1 1 1 1 ... 1 1 1

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0
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Definition

A pushdown automaton is a 6-tuple
 $(Q, \Sigma, \Gamma, \delta, q_0, F)$

- Q is the set of states
- Σ is the input alphabet
- Γ is the stack alphabet
- δ is the transition function

$$\delta: Q \times \Sigma_{\epsilon} \times \Gamma_{\epsilon} \rightarrow P(Q \times \Gamma_{\epsilon})$$

$$\Sigma_{\epsilon} = \Sigma \cup \{\epsilon\} \quad \Gamma_{\epsilon} = \Gamma \cup \{\epsilon\}$$

- q_0 is the start state
- $F \subseteq Q$ is the set of accept states.

A pushdown automaton $M = (Q, \Sigma, \Gamma, \delta, q_0, F)$ accepts input w if w can be written as $w = w_1 w_2 w_3 \dots w_m$, where $w_i \in \Sigma_\epsilon$ and sequence of states $r_0, r_1, r_2, \dots, r_m \in Q$ and strings $s_0, s_1, \dots, s_m \in \Gamma^*$ exist that satisfy the following three conditions.

The strings s_i represent the sequence of stack contents that M has on the accepting branch of computation.

1. $r_0 = q_0$ and $s_0 = \epsilon$. (This condition signifies that M starts properly, in the start state and with an empty stack.)
2. For $i = 0, \dots, m-1$, we have $(r_{i+1}, b) \in \delta(r_i, w_{i+1}, a)$ where $s_i = at$ and $s_{i+1} = bt$ for some $a, b \in \Gamma_\epsilon$ and $t \in \Gamma^*$. This condition states that M moves properly according to the state, stack, and next input symbol.
3. $r_m \in F$. This condition states that an accept state occurs at the input end.